

● RAG Learning

3.1 Ratio & Proportion

Pearson Edexcel GCSE Mathematics 1MA1 — Ratio, proportion and rates of change — spec refs R2–R8, R10, R12, R13 — Papers 1 (non-calculator), 2 and 3 (calculator)

3.1.1 Ratio Notation and Simplest Form (R4)

A **ratio** $a : b$ tells you how two (or more) quantities compare: for every a of the first there are b of the second. The order of the parts must match the order of the words — adults : children = $2 : 7$ is a different statement from children : adults = $2 : 7$.

A ratio is in its **simplest form** when all the parts are whole numbers with no common factor except 1. Divide every part by the highest common factor (HCF), or keep dividing by common factors until none are left.

- **Units first:** both parts must be in the same unit, e.g. $1.8 \text{ kg} : 600 \text{ g} = 1800 : 600$.
- **Decimals:** multiply every part by 10 or 100 to make whole numbers, e.g. $0.6 : 2.4 = 6 : 24 = 1 : 4$.
- **Fractions:** multiply every part by the lowest common denominator, e.g. $\frac{3}{4} : \frac{5}{6} = 9 : 10$ (multiply by 12).

Worked Example: Write $1.8 \text{ kg} : 600 \text{ g}$ in its simplest form.

In grams: $1800 : 600$. The HCF is 600, so the ratio is **$3 : 1$** .

Worked Example: Write $24 : 40 : 56$ in its simplest form.

Every part is a multiple of 8: $24 : 40 : 56 = \mathbf{3 : 5 : 7}$.

Writing a ratio in the form $1 : n$ or $n : 1$

Divide every part by the part that must become 1. Here n can be a decimal — the form $1 : n$ is used to compare ratios quickly (for example, map scales or the strength of two mixtures).

Worked Example: Write $4 : 9$ in the form $1 : n$.

Divide both parts by 4: $1 : 2.25$.

Exam Tip: If the question says “simplest form”, an answer such as $1 : 2.25$ will not score — give $4 : 9$. Only use decimals when the question asks for the form $1 : n$.

3.1.2 Fractions, Ratios and Linear Functions (R3, R6, R8)

Expressing one quantity as a fraction of another (R3)

Write the first quantity over the second, in the same units, and cancel. If the first quantity is larger, the fraction is **greater than 1** — that is fine.

Worked Example: (a) Write 21 as a fraction of 56: $\frac{21}{56} = \frac{3}{8}$.

(b) Write 75 cm as a fraction of 30 cm: $\frac{75}{30} = \frac{5}{2}$, so the first length is $2\frac{1}{2}$ times the second.

From a ratio to fractions (R6, R8)

Suppose adults : children on a trip = $2 : 7$. There are $2 + 7 = 9$ parts in total, so:

- $\frac{2}{9}$ of the group are adults and $\frac{7}{9}$ are children (part : whole);
- the number of adults is $\frac{2}{7}$ of the number of children, and the number of children is $\frac{7}{2}$ times the number of adults (part : part).

Multiplicative relationship

If $A : B = a : b$, then $A = \frac{a}{b} \times B$, and A makes up $\frac{a}{a+b}$ of the total.

Ratios and straight-line graphs (R8)

If $x : y = 2 : 5$ then $\frac{y}{x} = \frac{5}{2}$, so $y = 2.5x$. Plotting every pair in this ratio gives a **straight line through the origin** with gradient 2.5. A fixed ratio is the same thing as direct proportion (section 3.1.5).

Exam Tip: A very common error: with adults : children = 2 : 7, the fraction of adults is $\frac{2}{9}$, not $\frac{2}{7}$. Add the parts to get the whole.

3.1.3 Sharing in a Ratio and Ratio Problems (R5, R7)

Dividing a quantity in the ratio $a : b : c$

1. Add the parts: $a + b + c$
2. Find one part: quantity $\div$ total number of parts
3. Multiply one part by each number in the ratio

Worked Example: Share £540 in the ratio 5 : 7.

$5 + 7 = 12$ parts, so one part = $540 \div 12 = £45$.

Shares: $5 \times 45 = \textbf{£225}$ and $7 \times 45 = \textbf{£315}$. Check: $225 + 315 = 540$. ✓

Given one share or the difference

When the question gives you a **share** or the **difference between shares** rather than the total, use that to find one part.

Girls



Boys



5 extra parts = 35 more boys

Worked Example: In a club, girls : boys = 4 : 9. There are 35 more boys than girls. How many members are there?

Difference = $9 - 4 = 5$ parts, so 5 parts = 35 and 1 part = 7.

Total = 13 parts = $13 \times 7 = \mathbf{91}$ members.

Combining two ratios

Worked Example: $A : B = 5 : 6$ and $B : C = 4 : 3$. Find $A : B : C$.

The LCM of the two B values (6 and 4) is 12.

$A : B = 10 : 12$ and $B : C = 12 : 9$, so $A : B : C = \mathbf{10 : 12 : 9}$.

Proportion as equality of ratios (R7) — changing ratios

Two ratios are **in proportion** when they simplify to the same ratio. When a ratio changes because items are added or removed, write the original amounts as $2x$, $5x$, ... and set the new ratios equal.

Worked Example: A box holds red and green pens in the ratio 2 : 5. Nine red pens are added and the ratio becomes 1 : 2. How many pens were in the box at first?

Red = $2x$, green = $5x$. After: $(2x + 9) : 5x = 1 : 2$, so $2(2x + 9) = 5x$.

$4x + 18 = 5x \Rightarrow x = 18$. Original total = $7x = \mathbf{126}$ pens. (Check: $45 : 90 = 1 : 2 \checkmark$)

Real contexts: mixing, recipes and limiting quantities

When several ingredients are needed, find how much each one would allow you to make; the **smallest** answer is the most you can make.

Worked Example: Mortar uses cement : lime : sand = 1 : 1 : 6 (8 parts). A builder has 30 kg cement, 25 kg lime and 180 kg sand.

Cement: 1 part = 30 kg $\Rightarrow$ 240 kg mortar. Lime: 1 part = 25 kg $\Rightarrow$ 200 kg. Sand: 1 part = 30 kg $\Rightarrow$ 240 kg.

Greatest amount = **200 kg** — the lime runs out first.

Exam Tip: Edexcel ratio questions often hide the key information in the wording ("3 times as many", "£120 more than"). Write the ratio down first, then decide whether you know the total, one share or a difference.

3.1.4 Unitary Method, Best Buys, Scale Factors and Maps (R2, R5)

Unitary method and best buys

Find the cost (or amount) for **one** unit, then scale up. To compare value, work out the **price per unit** (lowest wins) or the **amount per £1** (highest wins), and write a clear conclusion.

Worked Example: Pasta: 500 g for £1.15, 1.2 kg for £2.64, 3 kg for £6.75. Which is the best value?

Price per kg: $1.15 \div 0.5 = £2.30$, $2.64 \div 1.2 = £2.20$, $6.75 \div 3 = £2.25$.

The **1.2 kg** bag is the best value.

Worked Example: The exchange rate is £1 = €1.18. Convert €472 into pounds.

Euros to pounds means divide: $472 \div 1.18 = £400$.

Scale factors, scale drawings and maps (R2)

A scale of $1 : n$ means that 1 unit on the drawing or map stands for n of the **same unit** in real life. Real length = drawing length $\times n$; drawing length = real length $\div n$.

Scale	1 cm on the map represents
1 : 200	200 cm = 2 m
1 : 10 000	10 000 cm = 100 m
1 : 25 000	25 000 cm = 250 m
1 : 200 000	200 000 cm = 2 km

Worked Example: A map has a scale of 1 : 25 000. Two churches are 3.4 km apart. How far apart are they on the map?

3.4 km = 340 000 cm. Map distance = $340\,000 \div 25\,000 = 13.6$ cm.

Worked Example: A plan of a car park uses a scale of 1 : 400. On the plan the car park is a rectangle 7 cm by 3.5 cm. Work out the real area.

Real lengths: $7 \times 400 = 2800$ cm = 28 m and $3.5 \times 400 = 1400$ cm = 14 m.

Area = $28 \times 14 = 392$ m². (Scale the lengths, not the area.)

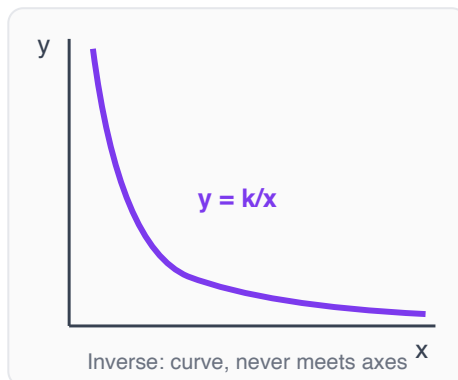
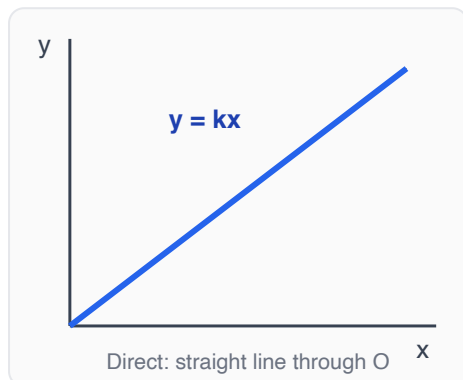
Exam Tip: Scale drawings often appear with bearings and loci on Edexcel papers. Convert carefully: 1 m = 100 cm and 1 km = 100 000 cm.

3.1.5 Direct and Inverse Proportion (R10, R13)

y is **directly proportional** to x ($y \propto x$) when $y = kx$ for some constant k : multiplying x by any number multiplies y by the same number.

y is **inversely proportional** to x when $y \propto \frac{1}{x}$, i.e. $y = \frac{k}{x}$ or $xy = k$: doubling x halves y .

(R13: " y is inversely proportional to x " means exactly the same as " y is proportional to $\frac{1}{x}$ ".)



Numerical and graphical problems

Worked Example: Direct: y is directly proportional to x , and $y = 27$ when $x = 6$. Find y when $x = 14$.

$\frac{y}{x}$ is constant: $27 \div 6 = 4.5$, so $y = 4.5x$. When $x = 14$: $y = \mathbf{63}$.

Worked Example: Inverse: 4 identical machines complete an order in 18 hours. How long would 6 machines take?

Total work = $4 \times 18 = 72$ machine-hours. Time = $72 \div 6 = \mathbf{12}$ hours.

To finish in 8 hours you would need $72 \div 8 = \mathbf{9}$ machines.

	Direct proportion	Inverse proportion
Notation	$y \propto x$	$y \propto \frac{1}{x}$
Equation	$y = kx$	$y = \frac{k}{x}$

Constant	$y \div x$	$x \times y$
x is trebled	y is trebled	y is divided by 3
Graph	straight line through the origin	reciprocal curve

Exam Tip: Before calculating, decide the type: "more machines $\rightarrow$ less time" is inverse; "more litres $\rightarrow$ more cost" is direct. Inverse proportion questions on Foundation papers are usually about workers, machines or speed and time.

3.1.6 Constructing Proportion Equations (R13) Higher

Higher tier students must **construct** (not just interpret) equations for proportion involving squares, cubes and square roots, then use them.

Four steps

1. Turn the words into an equation with k , e.g. $y = kx^2$, $d = \frac{k}{\sqrt{t}}$, $F = \frac{k}{d^2}$
2. Substitute the pair of values you are given
3. Solve for k and write out the full formula
4. Substitute again to answer the question

Statement	Equation	If x is multiplied by 3, y is...
y is proportional to x^2	$y = kx^2$	multiplied by 9
y is proportional to x^3	$y = kx^3$	multiplied by 27
y is proportional to $\sqrt{x}$	$y = k\sqrt{x}$	multiplied by $\sqrt{3}$
y is inversely proportional to x^2	$y = \frac{k}{x^2}$	divided by 9

y is inversely proportional to $\sqrt{x}$

$$y = \frac{k}{\sqrt{x}}$$

divided by $\sqrt{3}$

Worked Example: y is directly proportional to x^2 . When $x = 3$, $y = 36$. Find y when $x = 2.5$.

$$y = kx^2 \Rightarrow 36 = 9k \Rightarrow k = 4, \text{ so } y = 4x^2.$$

$$y = 4 \times 2.5^2 = 4 \times 6.25 = \mathbf{25}.$$

Worked Example: The force F is inversely proportional to the square of the distance d . $F = 50$ when $d = 2$. Find d when $F = 8$.

$$F = \frac{k}{d^2} \Rightarrow 50 = \frac{k}{4} \Rightarrow k = 200, \text{ so } F = \frac{200}{d^2}.$$

$$8 = \frac{200}{d^2} \Rightarrow d^2 = 25 \Rightarrow d = \mathbf{5}.$$

Worked Example: T is inversely proportional to $\sqrt{L}$. $T = 9$ when $L = 4$. Find T when $L = 81$.

$$T = \frac{k}{\sqrt{L}} \Rightarrow 9 = \frac{k}{2} \Rightarrow k = 18. \text{ When } L = 81: T = \frac{18}{9} = \mathbf{2}.$$

Percentage changes

If $y \propto x^2$ and x increases by 30%, then y is multiplied by $1.3^2 = 1.69$ — an increase of **69%**, not 60%.

Exam Tip: Edexcel mark schemes usually award a mark for the formula with k replaced by its value (e.g. " $y = 4x^2$ "). Write it on its own line, even if the question does not ask for it.

3.1.7 Comparing Lengths, Areas and Volumes (R12)

Ratio notation can be used to compare lengths, areas and volumes. When one shape is an enlargement of another, all corresponding lengths are in the same ratio — this is the link between ratio, **scale factors** and **similarity**. It also explains the **trigonometric ratios**: in similar right-angled triangles the ratio $\frac{\text{opposite}}{\text{hypotenuse}}$ is the same for the same angle.

Worked Example: Rectangle P is 2 cm by 5 cm. Rectangle Q is 6 cm by 15 cm.

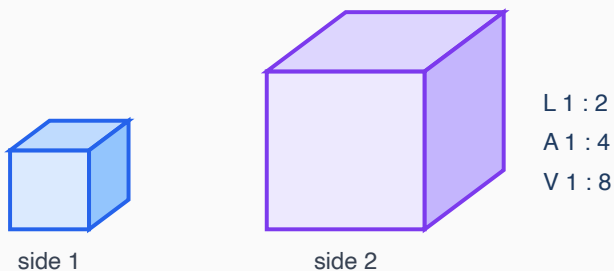
Length ratio = $2 : 6 = 1 : 3$. Area ratio = $10 : 90 = 1 : 9$, which is $1^2 : 3^2$.

Area and volume of similar shapes (G19) Higher

If the length scale factor is k (lengths in the ratio $a : b$):

area scale factor = k^2 (areas in the ratio $a^2 : b^2$)

volume scale factor = k^3 (volumes in the ratio $a^3 : b^3$)



Worked Example: Two similar tins are 8 cm and 12 cm tall. The smaller tin has volume 320 cm^3 and surface area 200 cm^2 .

$$k = 12 \div 8 = 1.5.$$

$$\text{Volume of larger tin} = 320 \times 1.5^3 = 320 \times 3.375 = 1080 \text{ cm}^3.$$

$$\text{Surface area of larger tin} = 200 \times 1.5^2 = 200 \times 2.25 = 450 \text{ cm}^2.$$

Worked Example: Two similar statues have volumes 16 cm^3 and 250 cm^3 . Find the ratio of their heights.

Volume ratio $16 : 250 = 8 : 125$. Length ratio $= \sqrt[3]{8} : \sqrt[3]{125} = 2 : 5$.

Exam Tip: Working backwards from areas or volumes, square-root or cube-root the scale factor to get the length scale factor *before* doing anything else.

Summary

Skill	Key idea
Simplify a ratio (R4)	Same units, then divide by the HCF
Fraction of another quantity (R3)	$\frac{\text{first}}{\text{second}}$, can be greater than 1
Share in a ratio (R5)	Total parts $\rightarrow$ one part $\rightarrow$ each share
Ratio to fraction (R6, R8)	$a : b$ gives $\frac{a}{a+b}$ of the whole
Best buy	Compare price per unit
Scale $1 : n$ (R2)	Real = map $\times n$, then convert units
Direct proportion (R10)	$y = kx$; straight line through the origin
Inverse proportion (R10, R13)	$y = \frac{k}{x}$; xy constant
Proportion equations (R13) Higher	Write with k , substitute, find k , use formula
Similar shapes (R12, G19) Higher	Lengths k , areas k^2 , volumes k^3